Chapter 1

TT

1.1 SYS-0

The alphabet is listed as follows.

- $V = \{v_i\}_{i\geq 0} \cup \{A_i\}_{i\geq 0}$;
- C is the set of constant symbols;
- $B = \{\Pi, \lambda, \Sigma, \cdots\}$ is the set of binders;
- $U = \{\mathrm{Type}_i\}_{i\geq 0}$ is the set of universes;
- $P = \{(,), :, \equiv\}$ is the set of punctuation.

All of these sets are at most countable.

The whole formulas we use is contained by set

$$\mathrm{F} = V|C|U|B(V:\mathrm{F})\mathrm{F}|(\mathrm{FF})|(C\mathrm{F}^*).$$

Notation 1.1.1 *Here we announce some commonly used notations for avoiding ambiguity.*

- $\boldsymbol{a} = a_0 a_1 \cdots a_m$;
- $f(\boldsymbol{x}) := f$ *where* $f \in \mathrm{F}$ *and* $\mathrm{Fr}\, f \subset \{x_0, \cdots, x_m\}$;
- $(f\boldsymbol{x})$ *refers* $(\cdots((fx_0)x_1)\cdots)$, *note that this may differ the notation* $(fx_0x_1\cdots x_m)$, *which is equal to our first definition;*
- *for any* $b \in B$, $b(\boldsymbol{v}:\boldsymbol{A})N := b(v_0:A_0)b(v_1:A_1)\cdots b(v_m:A_m)N$;
- $\boldsymbol{v}:\boldsymbol{A}$ *refers*
$$v_0 : A_0 \quad v_1 : A_1 \quad \cdots \quad v_s : A_s;$$
- $M \to N := \Pi(v:M)N$, *in which* $v \notin \mathrm{Fr}\, N$;
- *to distinguish between terms in* F *and constants in* C, *we use a mono font for constants;*
- $\mathcal{U} := \mathrm{Type}_{\aleph_0}$, *and* $M : \mathcal{U}$ *implies there exists a* $k < \aleph_0$ *such that* $M : \mathrm{Type}_k$.
- *for any relation* $\mathcal{R} \subset \mathrm{F}^2$, $\mathrm{Compatible}\mathcal{R}$ *is a abbrivation of the statement: for all if* $M\mathcal{R}N$, *then for any* $L, A, \phi_1, \cdots, \phi_s \in \mathrm{F}$, $b \in B$, $\mathtt{c} \in C$ *and* $v \in V$, $(ML)\mathcal{R}(NL)$, $(LM)\mathcal{R}(LN)$, $b(v:A)M\mathcal{R}b(v:A)N$ *and*
$$(\mathtt{c}\phi_1\cdots\phi_k M\phi_{k+1}\cdots\phi_s)\mathcal{R}(\mathtt{c}\phi_1\cdots\phi_k N\phi_{k+1}\cdots\phi_s)$$

Definition 1.1.2 *A definition D has the form*

$$\mathtt{c}(\boldsymbol{v} : \boldsymbol{A}) \equiv d(\boldsymbol{v}) : B,$$

where $v \in V$*,* $\mathtt{c} \in C$ *and all of* $A, B, d(\boldsymbol{v})$ *are in* F*.*

Definition 1.1.3 *A sequent has the form*

$$\boldsymbol{D} \quad \boldsymbol{v} : \boldsymbol{A} \quad M : A$$

where $M : A$ *is called a judgement, and both* M *and* A *are in* F*. The set of all sequents is denoted by* Sqt*.*

Legal judgements consist a subset of F : F, which is derived from a series of **rules**. In our SYS-0, we first introduce the following rules, each of them can be viewed as a relation on $\mathrm{Sqt}^* \times \mathrm{Sqt}$.

Sort

If $m < n$, then

$$\_ \quad \_ \quad \mathrm{Type}_m : \mathrm{Type}_n$$

Var

If $v \in V \setminus (\Gamma \cap V)$, then

$$\boldsymbol{D} \quad \Gamma \quad A : \mathcal{U}$$
$$\boldsymbol{D} \quad \Gamma \quad v : A \quad v : A$$

Weak

If $v \in V \setminus (\Gamma \cap V)$, then

$$\boldsymbol{D} \quad \Gamma \quad M : A$$
$$\boldsymbol{D} \quad \Gamma \quad B : \mathcal{U}$$
$$\boldsymbol{D} \quad \Gamma \quad v : B \quad M : A$$

Compu

Given the equivalence relation $=_a \subset \mathrm{F}^2$, if $B =_a B'$, then

$$\boldsymbol{D} \quad \Gamma \quad M : B$$
$$\boldsymbol{D} \quad \Gamma \quad B' : \mathcal{U}$$
$$\boldsymbol{D} \quad \Gamma \quad M : B'$$

1.2 SYS-1

我们通过引入新符号, 增加新规则和关系来增加新的项和类型, 这个过程大致遵循一个固定的工作流: 1. Formation: 引入新规则, 让系统在指定条件下可以导出新类型; 2. Constuction: 引入新规则, 让系统在指定条件下可以导出新项, 并且这些项都具有新类型; 3. Elimination: 引入新规则, 告知系统新项应该如何使用; 4. Computaiton: 引入 F 上的新关系, 告知系统新系统中某些表达和旧系统中的某些表达具有同样的语义.

现在我们将拓展 SYS-0, 增加 λ-Terms 和 Π-Types.

SYS-1: Form

If $s_1 \leq s_2$, then

$$\boldsymbol{D} \quad \Gamma \quad A : \mathcal{U}$$
$$\boldsymbol{D} \quad \Gamma \quad v : A \qquad\qquad B : \mathcal{U}$$
$$\boldsymbol{D} \quad \Gamma \quad \Pi(v : A)B : \mathcal{U}$$

SYS-1: Constr

$$\begin{array}{lll} \boldsymbol{D} & \Gamma & v : A \qquad\qquad\qquad\qquad M : B \\ \boldsymbol{D} & \Gamma & \Pi(v : A)B : \mathcal{U} \\ \boldsymbol{D} & \Gamma & \lambda(v : A)M : \Pi(v : A)B \end{array}$$

SYS-1: Elim

$$\begin{array}{lll} \boldsymbol{D} & \Gamma & M : \Pi(v : A)B \\ \boldsymbol{D} & \Gamma & N : A \\ \boldsymbol{D} & \Gamma & (MN) : B\frac{N}{v} \end{array}$$

SYS-1: Compu

Definition 1.2.1 $\to_0 \subset \mathrm{F}^2$ *is the samllest relation satisfies the following properties:*

- $\diamond$ *for any* A, M *in* F, $b \in B$ *and* $v_0, v_1 \in V$ *such that* $v_1 \in (M \cap V) \setminus \{v_0\}$

$$b(v_0 : A)M \to_0 b(v_1 : A)M\frac{v_1}{v_0};$$

- $\diamond$ Compatible $\to_0$.

Definition 1.2.2 $\to_1 \subset \mathrm{F}^2$ *is the samllest relation satisfies the following properties:*

- $\diamond$ *for any* A, M, N *in* F, $(\lambda(v : A)MN) \to_1 M\frac{N}{v}$*;*
- $\diamond$ Compatible $\to_1$.

$=_1$ *is the equivalence genertated by* $\bigcup\{\to_i\}_{i=0}^{1}$.

1.3 SYS-2

在 SYS-1 的基础上加入 Σ-Types 得到 SYS-2.

SYS-2: Form

$$\begin{array}{lll} \boldsymbol{D} & \Gamma & A : \mathcal{U} \\ \boldsymbol{D} & \Gamma & v : A \qquad\qquad B : \mathcal{U} \\ \boldsymbol{D} & \Gamma & \Sigma(v : A)B \end{array}$$

SYS-2: Constr

$$\begin{array}{lll} \boldsymbol{D} & \Gamma & \Sigma(v : A)B \\ \boldsymbol{D} & \Gamma & a : A \\ \boldsymbol{D} & \Gamma & b : B\frac{a}{v} \\ \boldsymbol{D} & \Gamma & (\mathtt{pair}ab) : \Sigma(v : A)B \end{array}$$

定义 Elim 前先明确我需求, 对任何 Type 或者 Universe Q, 通过给定的项 $\phi : \Pi(v : A)(Bv \to Q)$, 其中 Bv 指的是公式的合并 (Bv), 我们可以构造出一个具有类型 $(\Sigma(v : A)Bv) \to Q$ 的项 ψ. 所以, 有理由将规则设计成下面这个样子:

$$\begin{array}{lll} \boldsymbol{D} & \Gamma & \Sigma(v : A)B : \mathcal{U} \\ \boldsymbol{D} & \Gamma & \phi : \Pi(v : A)(Bv \to Q) \\ \boldsymbol{D} & \Gamma & (\mathtt{Elim}\phi Q) : (\Sigma(v : A)Bv) \to Q \end{array}$$

然后再加入新关系 $\to_2$, 其包含下面内容

$$((\mathtt{Elim}\phi Q)(\mathtt{pair}ab)) \to_2 ((\phi a)b)$$

除此之外还有一种更普遍的方法, 它是从更复杂的功能中下放到这里的.

SYS-2: Elim

$$\begin{array}{lll} \boldsymbol{D} & \Gamma & \Sigma(v:A)B:\mathcal{U} \\ \boldsymbol{D} & \Gamma & \begin{array}{c}(\mathtt{Ind}\Sigma(v:A)B):\Pi(X:(\Sigma(v_0:A)B)\to\mathcal{U}).\\ {[\Pi(v_1:A)(v_2:Bv_2)(X\mathtt{pair}v_1v_2)\to\Pi(v_3:\Sigma(v_4:A)Bv_4)(Xv_3)]}\end{array} \end{array}$$

可见如果给出常函数 $H=\lambda(X:\Sigma(v_0:A)B)Q:\Pi(X:(\Sigma(v_0:A)B)\to\mathcal{U}$, 还有

$$\phi:\Pi(v_0:A)(v_1:Bv_0)H(\mathtt{pair}v_0v_1)=\Pi(v_0:A)(v_1:Bv_0)Q,$$

那么

$$(\mathtt{Ind}\Sigma(v:A)B)(H\phi):(\Sigma(v:A)Bv)\to Q.$$

SYS-2: Compu

Definition 1.3.1 $\to_2\subset \mathrm{F}^2$ *is the samllest relation satisfies the following properties:*

- ⋄ *for all* $a,b,A,B,\phi,H\in\mathrm{F}$,
$$(\mathtt{Ind}\Sigma(v:A)B)(H\phi ab)\to_2(\phi a)b$$
- ⋄ Compatible $\to_2$.

$=_2$ *is the equivalence genertated by* $\bigcup\{\to_i\}_{i=0}^2$.

1.4 SYS-3

在 SYS-2 中加入自然数类型 $\mathbb{N}$ 使其成为 SYS-3.

SYS-3: Form

$$\begin{array}{lll} \boldsymbol{D} & \Gamma & \_ \\ \boldsymbol{D} & \Gamma & \mathbb{N}:\mathcal{U} \end{array}$$

SYS-3: Constr

$$\begin{array}{lll} \boldsymbol{D} & \Gamma & \mathbb{N}:\mathcal{U} \\ \boldsymbol{D} & \Gamma & \mathtt{0}:\mathbb{N} \end{array}$$

and

$$\begin{array}{lll} \boldsymbol{D} & \Gamma & \mathbb{N}:\mathcal{U} \\ \boldsymbol{D} & \Gamma & \mathtt{S}:\mathbb{N}\to\mathbb{N} \end{array}$$

SYS-3: Elim

$$\begin{array}{lll} \boldsymbol{D} & \Gamma & \mathbb{N}:\mathcal{U} \\ \boldsymbol{D} & \Gamma & (\mathtt{Ind}\mathbb{N}):\Pi(X:\mathbb{N}\to\mathcal{U})[(X\mathtt{0})\to\Pi(v_0:\mathbb{N})((Xv_0)\to X(\mathtt{S}v_0))\to\Pi(v_1:\mathbb{N})(Xv_1)] \end{array}$$

Example 1.4.1 *Let* P *be* $\lambda(v:\mathbb{N})\mathrm{Type}_0$, *which implies that* P *is a property on* $\mathbb{N}$. *In addition, assume that* p_0 *and* p_S *are two legal elements in* F, *satisfy* $p_0:P\mathtt{0}$ *and* $p_S:\Pi(v:\mathbb{N})((Pv)\to P(\mathtt{S}v))$, *means that both propositions* $P\mathtt{0}$ *and* $\forall n:\mathbb{N}\ Pn\to P\mathtt{S}n$ *are ture, then*

$$(\mathtt{Ind}\mathbb{N})(Pp_0p_S):\Pi(v:\mathbb{N})(Pv),$$

namely, the proposition $\forall n:\mathbb{N}\ P(n)$ *is true.*

SYS-3: Compu

Definition 1.4.2 $\to_3 \subset \mathrm{F}^2$ *is the samllest relation satisfies the following properties:*

- *for all* $\phi, a, b, c \in \mathrm{F}$,

$$(\mathtt{Ind}\mathbb{N})(\phi ab\mathtt{0}) \to_3 a$$

and

$$(\mathtt{Ind}\mathbb{N})(\phi ab(\mathtt{S}c)) \to_3 (bc)((\mathtt{Ind}\mathbb{N})(\phi abc))$$

- Compatible $\to_3$.

$=_3$ *is the equivalence genertated by* $\bigcup\{\to_i\}_{i=0}^{3}$.

1.5 CIC

Definition 1.5.1

- *Arity is genertated by the syntax below:*

$$\mathbb{A} = U|(V : \mathrm{F})\mathbb{A}.$$

- *The Strictly Positive Types of* ω *is genertated by*

$$\mathbb{P}(\omega) = \omega$$